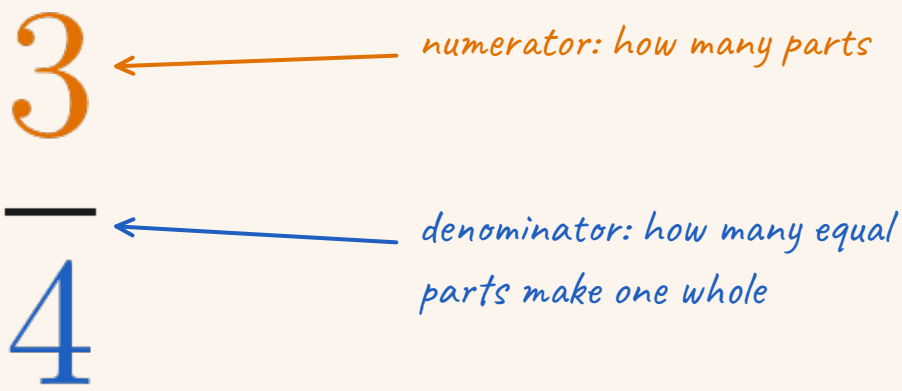
Chapter 2

Fractions

the rules for each operation, and why they work



Everything you do with two fractions comes from **three formulas**: one for adding and subtracting, one for multiplying, one for dividing. They are on the next pages. Put your numbers in for a, b, c, d and work it out; the rest of the chapter is examples.

Goes with **Lecture 1** (Combining fractions). Uses factors, HCF and LCM from Chapter 1.

What a fraction means

DEFINITION: reading a fraction

$\frac{a}{b}$ means: split one whole into **b equal parts** and take **a** of them.

a is the **numerator** (top). **b** is the **denominator** (bottom). b can never be 0.

The fraction line **is** a division: $\frac{a}{b} = a \div b$. So $\frac{12}{3} = 4$ and $\frac{3}{4} = 0.75$.

Any whole number can be written over 1: $5 = \frac{5}{1}$.



$$\frac{3}{4}$$

3 of 4 equal parts

The parts must be **equal**. The denominator gives the parts their name: halves, thirds, quarters, fifths... The numerator counts how many of those parts you have.

- **Proper fraction**: top smaller than bottom, so less than 1 (e.g. $\frac{3}{4}$).
- **Improper fraction**: top bigger than bottom, so more than 1 (e.g. $\frac{7}{4}$).
- **Mixed number**: a whole number plus a proper fraction (e.g. $1\frac{3}{4}$, "one and three quarters").

The three formulas

These are the whole lesson. Put your numbers in for a, b, c, d and work it out.

THE RULES

Add / subtract

same for + and −

$$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$$

Multiply

multiply across

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

Divide

flip the 2nd, then multiply across

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

With numbers ($a = 2$, $b = 3$, $c = 4$, $d = 5$: Lecture 1's examples):

$$\frac{2}{3} + \frac{4}{5} = \frac{2 \times 5 + 3 \times 4}{3 \times 5} = \frac{22}{15}$$ ← *ad + bc over bd*

$$\frac{2}{3} - \frac{4}{5} = \frac{2 \times 5 - 3 \times 4}{3 \times 5} = -\frac{2}{15}$$ ← *ad − bc over bd*

$$\frac{2}{3} \times \frac{4}{5} = \frac{2 \times 4}{3 \times 5} = \frac{8}{15}$$ ← *ac over bd*

$$\frac{2}{3} \div \frac{4}{5} = \frac{2 \times 5}{3 \times 4} = \frac{10}{12}$$ ← *ad over bc*

INPUTS: getting the numbers into a/b form

A whole number goes over 1: $4 = \frac{4}{1}$.

A mixed number becomes an improper fraction first: $2\frac{1}{3} = \frac{2\times3+1}{3} = \frac{7}{3}$.

Simplifying the answer is an optional tidy-up: $\frac{10}{12}$ and $\frac{5}{6}$ are the same number. Do it when a question asks for the simplest form.

Where the add/subtract formula comes from

Multiplying a fraction's top and bottom by the same number doesn't change its value. So multiply the first fraction by $\frac{d}{d}$ and the second by $\frac{b}{b}$:

$$\frac{a}{b} + \frac{c}{d} = \frac{ad}{bd} + \frac{bc}{bd} = \frac{ad + bc}{bd}$$

Both fractions are now over bd, so their tops add. That's the whole derivation. When a textbook says "find a common denominator", it means this step; the formula has already done it for you with bd.

Multiplying needs no such step: $\frac{a}{b} \times \frac{c}{d}$ is just $a \times c \div (b \times d)$. Dividing by $\frac{c}{d}$ is multiplying by $\frac{d}{c}$, because $\frac{c}{d} \times \frac{d}{c} = 1$.

Adding and subtracting

$$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$$

Example: $\frac{3}{4} + \frac{1}{6}$.

$$= \frac{3 \times 6 + 4 \times 1}{4 \times 6} = \frac{18 + 4}{24} = \frac{22}{24}$$

$a = 3, b = 4, c = 1, d = 6$

$$= \frac{11}{12}$$ ← optional: simplify, ÷ 2

Example: $\frac{5}{6} - \frac{1}{4}$.

$$= \frac{5 \times 4 - 6 \times 1}{6 \times 4} = \frac{20 - 6}{24} = \frac{14}{24} = \frac{7}{12}$$

$ad - bc$ over bd

Example: the bottoms are already the same, $\frac{1}{5} + \frac{2}{5}$.

$$= \frac{1 \times 5 + 5 \times 2}{5 \times 5} = \frac{15}{25} = \frac{3}{5}$$ ← the formula still works

Shortcut when $b = d$: the formula's answer always simplifies to "add the tops, keep the bottom", $\frac{1+2}{5} = \frac{3}{5}$. Use it or ignore it; the answer is the same.

Optional, for smaller numbers: instead of bd you may use any number both bottoms divide into, such as the lowest common multiple. For $\frac{3}{4} + \frac{1}{6}$ that is 12: $\frac{9}{12} + \frac{2}{12} = \frac{11}{12}$, the same answer as above without the simplifying at the end.

Slide error alert. On the Lecture 1 slide for **subtraction**, the left-hand side is printed with a **+** sign. The correct formula has a minus on both sides.

Sense check (your lecturer's habit): $\frac{2}{3}$ and $\frac{4}{5}$ are each a bit under 1, so $\frac{2}{3}+\frac{4}{5}=\frac{22}{15}=1\frac{7}{15}$, a bit under 2, is sensible. ✓

Negative fractions

The lecture's subtraction example is $\frac{2}{3}-\frac{4}{5}$, which comes out **negative**: you take away more than you started with.

RULE: negative fractions

The minus sign can sit in front, on top, or on the bottom: $-\frac{a}{b}=\frac{-a}{b}=\frac{a}{-b}$. Put it on **top** while you work.

The three formulas work unchanged; the tops follow the rules for negative numbers (Chapter 1).

Multiplying and dividing: same sign gives **+**, different signs give **−**.

Two minus signs cancel: $\frac{-a}{-b}=\frac{a}{b}$.

$$\frac{2}{3}-\frac{4}{5}=\frac{2\times5-3\times4}{3\times5}$$ ← ad − bc over bd

$$=\frac{10-12}{15}=-\frac{2}{15}$$ ← 10 − 12 = −2

Sense check: you took away $\frac{4}{5}=0.8$ from $\frac{2}{3}\approx0.67$, so the answer must be a little below zero. $-\frac{2}{15}\approx-0.13$ ✓

$$\frac{-1}{4}+\frac{2}{3}=\frac{-1\times3+4\times2}{4\times3}=\frac{5}{12}$$

−3 + 8 = 5

$$\left(-\frac{2}{3}\right)\times\frac{9}{4}=-\frac{18}{12}=-\frac{3}{2}$$ ← different signs: negative

$$\left(-\frac{1}{2}\right) \div \left(-\frac{1}{4}\right) = \frac{1}{2} \times \frac{4}{1} = 2$$ ← same signs: positive

Multiplying

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

Multiply across. Whole numbers go over 1; mixed numbers become improper fractions first. "$\times$" with fractions means "of": $\frac{2}{3} \times \frac{4}{5}$ is two thirds of four fifths.

Example: Lecture 1, $\frac{2}{3} \times \frac{4}{5}$.

$$\frac{2}{3} \times \frac{4}{5} = \frac{2 \times 4}{3 \times 5} = \frac{8}{15}$$ ← ac over bd

Optional shortcut (cross-cancelling): before multiplying you may divide any top and any bottom by a common factor, to keep the numbers small. Only across $\times$, never $+$ or $-$.

Example: cross-cancelling, $\frac{3}{4} \times \frac{2}{9}$.

$$\frac{\cancel{3}^{1}}{\cancel{4}_{2}} \times \frac{\cancel{2}^{1}}{\cancel{9}_{3}} = \frac{1 \times 1}{2 \times 3} = \frac{1}{6}$$ ← 3 with 9 (÷3); 2 with 4 (÷2)

Example: a whole number, $3 \times \frac{2}{5}$.

$$\frac{3}{1} \times \frac{2}{5} = \frac{6}{5} = 1\frac{1}{5}$$ ← 3 = 3/1

Example: mixed numbers, $1\frac{1}{2} \times 2\frac{1}{2}$.

$$= \frac{3}{2} \times \frac{5}{2} = \frac{15}{4} = 3\frac{3}{4}$$ ← change to improper first

Not $1 \times 2 + \frac{1}{2} \times \frac{1}{2} = 2\frac{1}{4}$. Multiplying the wholes and the parts separately misses the cross terms (like expanding brackets, Chapter 11).

RULE: a fraction of an amount

"$\frac{a}{b}$ of N" means $\frac{a}{b} \times N$: divide N by the bottom, then multiply by the top.

$$\frac{3}{5} \text{ of } 40 = \frac{40}{5} \times 3 = 8 \times 3 = 24$$

÷ bottom, × top

Dividing

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

Change ÷ to ×, flip the **second** fraction only, then multiply across. Whole numbers go over 1 ($4 = \frac{4}{1}$, flipped $\frac{1}{4}$); mixed numbers become improper fractions first.

The **reciprocal** of a fraction is the fraction flipped upside down: $\frac{2}{3} \to \frac{3}{2}$, $4 \to \frac{1}{4}$. A number times its reciprocal is always 1.

Example: Lecture 1, $\frac{2}{3} \div \frac{4}{5}$.

$$\frac{2}{3} \div \frac{4}{5} = \frac{2}{3} \times \frac{5}{4}$$ ← keep, change, flip

$$= \frac{10}{12} = \frac{5}{6}$$ ← multiply, then simplify ÷ 2

Example: dividing by a whole number, $\frac{2}{3} \div 4$.

$$\frac{2}{3} \div \frac{4}{1} = \frac{2}{3} \times \frac{1}{4} = \frac{2}{12} = \frac{1}{6}$$ ← 4 = 4/1, flipped 1/4

Example: mixed numbers, $2\frac{1}{2} \div 1\frac{1}{4}$.

$$= \frac{5}{2} \div \frac{5}{4} = \frac{5}{2} \times \frac{4}{5} = \frac{20}{10} = 2$$ ← improper first

Why flipping works: $\frac{c}{d} \times \frac{d}{c} = 1$, so multiplying by $\frac{d}{c}$ undoes multiplying by $\frac{c}{d}$, which is exactly what dividing by $\frac{c}{d}$ does. E.g. $3 \div \frac{1}{4} = 3 \times 4 = 12$: twelve quarters in 3.

Sense check: dividing by a number less than 1 makes the answer bigger; dividing by a number more than 1 makes it smaller.

Simplifying an answer (equivalent fractions)

The formulas often give answers like $\frac{10}{12}$. That is already correct; simplifying just writes the same number with smaller numbers.

RULE: equivalent fractions

Multiplying or dividing the top **and** the bottom by the **same** non-zero number gives the same value: $\frac{a}{b} = \frac{a \times k}{b \times k}$.

Adding or subtracting the same number to top and bottom does NOT keep the value.

$$\frac{3}{4} = \frac{3 \times 5}{4 \times 5} = \frac{15}{20}$$ ← top AND bottom × 5

$$\frac{1}{2} \neq \frac{1+1}{2+1} = \frac{2}{3}$$ ← adding 1 to both changes the value

PROCEDURE: simplifying (cancelling)

1. Find the **highest common factor** (HCF) of the top and bottom (Chapter 1).
2. Divide the top and the bottom by it.
3. Check: if no number except 1 divides both, the fraction is in **simplest form**. (Smaller steps are fine: divide by any common factor, repeat until none is left.)

Example: simplify $\frac{12}{18}$.

$$\mathrm{HCF}(12, 18) = 6$$ ← factors of 12: 1,2,3,4,6,12; of 18: 1,2,3,6,9,18

$$\frac{12 \div 6}{18 \div 6} = \frac{2}{3}$$ ← divide top and bottom by 6

Cancel factors, never terms. In $\frac{6}{8}$ you divide the **whole** top and the **whole** bottom. In $\frac{2+4}{2+6}$ you may **not** cross out the 2s: work out the top and bottom first ($\frac{6}{8}$), then cancel.

Mixed numbers and improper fractions

RULE: converting

Mixed to improper: whole × bottom, add the top, keep the bottom: $p\frac{a}{b} = \frac{p \times b + a}{b}$.

Improper to mixed: divide top by bottom. The quotient is the whole number, the remainder is the new top, the bottom stays the same.

$$2\frac{1}{3} = \frac{2 \times 3 + 1}{3} = \frac{7}{3}$$ ← 2 wholes = 6 thirds, plus 1 third

$$\frac{11}{4} : \; 11 \div 4 = 2 \text{ r } 3$$ ← 4 goes into 11 twice, 3 left over

$$\frac{11}{4} = 2\frac{3}{4}$$ ← 2 wholes and 3 quarters

Negative mixed numbers: $-1\frac{3}{4}$ means $-(1 + \frac{3}{4}) = -\frac{7}{4}$. The minus sign applies to the whole thing, **not** just the 1.

Mixed numbers

Change each mixed number to an improper fraction, then use the formula as normal.

Example: $1\frac{1}{2} + 2\frac{3}{4}$.

$$= \frac{3}{2} + \frac{11}{4} = \frac{3 \times 4 + 2 \times 11}{2 \times 4} = \frac{34}{8}$$

improper, then the formula

$$= \frac{17}{4} = 4\frac{1}{4}$$ ← *simplify: 17 ÷ 4 = 4 r 1*

Example: $3\frac{1}{4} - 1\frac{2}{3}$.

$$= \frac{13}{4} - \frac{5}{3} = \frac{13 \times 3 - 4 \times 5}{4 \times 3} = \frac{19}{12} = 1\frac{7}{12}$$

improper, then the formula

Comparing fractions

$$\frac{a}{b} > \frac{c}{d} \iff ad > bc \qquad (\text{positive } b,\ d)$$

Cross-multiply and compare ad with bc. It's the top of the subtraction formula: $\frac{a}{b} - \frac{c}{d} = \frac{ad-bc}{bd}$ is positive exactly when $ad > bc$. Or turn both into decimals (top ÷ bottom).

Example: Lecture 1, $\frac{9}{17}$ or $\frac{49}{97}$?

$$9 \times 97 = 873 \qquad 49 \times 17 = 833$$

each top × the OTHER bottom

$$873 > 833 \ \Rightarrow\ \frac{9}{17} > \frac{49}{97}$$

The lecture does it with decimals: $0.529 > 0.505$.

Order of operations and whole numbers

RULE: order of operations with fractions

The usual order applies (Chapter 1): **brackets, powers, × and ÷ (left to right), + and − (left to right)**.

A fraction bar groups like brackets: in $\frac{1+2}{3+4}$, work out the top and the bottom first.

"Of" means ×.

$$\frac{1}{2}+\frac{1}{3}\times\frac{3}{4}=\frac{1}{2}+\frac{1}{4}=\frac{3}{4}$$ ← × first, then +

$$\left(\frac{1}{2}+\frac{1}{3}\right)\times\frac{3}{4}=\frac{5}{6}\times\frac{3}{4}=\frac{15}{24}=\frac{5}{8}$$

brackets first

Whole numbers with fractions. Write the whole number over 1, then use the formula:

$$3+\frac{2}{5}=\frac{3}{1}+\frac{2}{5}=\frac{3\times5+1\times2}{1\times5}=\frac{17}{5}$$

3 = 3/1

$$2-\frac{3}{4}=\frac{2}{1}-\frac{3}{4}=\frac{2\times4-1\times3}{1\times4}=\frac{5}{4}$$

2 = 2/1

Fractions or decimals? $\frac{1}{3}=0.333...$ never ends, so any decimal you write is slightly wrong. For exact work (algebra), **keep fractions**. For measured data (statistics), decimals are fine (Lecture 1).

Find the missing number

PROCEDURE: finding an unknown fraction

1. Write the unknown as $?$ (or x).
2. **Undo** the operation that was done to it, using the opposite operation on the known values: + is undone by −, × is undone by ÷ (and the other way round).
3. Then use the three formulas to work out the answer.
4. Check by putting your answer back into the original question.

Example: $\frac{3}{4} = \frac{?}{20}$.

$$4 \times 5 = 20 \Rightarrow ? = 3 \times 5 = 15$$

← the bottom was × 5, so the top is × 5

Example: $\frac{2}{5} + ? = \frac{9}{10}$.

$$? = \frac{9}{10} - \frac{2}{5} = \frac{9 \times 5 - 10 \times 2}{10 \times 5} = \frac{25}{50} = \frac{1}{2}$$

undo + with −

Example: $? \times \frac{3}{4} = \frac{1}{2}$.

$$? = \frac{1}{2} \div \frac{3}{4} = \frac{1}{2} \times \frac{4}{3} = \frac{4}{6} = \frac{2}{3}$$

← undo × with ÷

Example: $\frac{5}{6} \div ? = \frac{5}{3}$.

$$? = \frac{5}{6} \div \frac{5}{3} = \frac{5}{6} \times \frac{3}{5} = \frac{15}{30} = \frac{1}{2}$$

← the unknown is the divisor: ? = 5/6 ÷ 5/3

Example: $\frac{3}{8}$ of a number is 15. What is the number?

$$\frac{3}{8} \times ? = 15 \Rightarrow ? = 15 \div \frac{3}{8} = 15 \times \frac{8}{3} = 40$$

undo × with ÷

The mistakes to avoid

✗ Common mistakes

✗ $\frac{1}{2} + \frac{1}{3} = \frac{2}{5}$

✓ $\frac{1}{2} + \frac{1}{3} = \frac{1 \times 3 + 2 \times 1}{2 \times 3} = \frac{5}{6}$

adding tops and adding bottoms isn't the formula: use (ad + bc)/bd

✗ $\frac{\cancel{3}}{4} + \frac{1}{\cancel{3}}$

✓ $\frac{3}{4} + \frac{1}{3} = \frac{9+4}{12} = \frac{13}{12}$

cross-cancelling only works across ×

✗ $\frac{2}{3} \div \frac{4}{5} = \frac{3}{2} \times \frac{4}{5}$

✓ $\frac{2}{3} \div \frac{4}{5} = \frac{2}{3} \times \frac{5}{4}$

flip the SECOND fraction only

✗ $1\frac{1}{2} \times 2\frac{1}{2} = 2\frac{1}{4}$

✓ $\frac{3}{2} \times \frac{5}{2} = \frac{15}{4} = 3\frac{3}{4}$

change mixed numbers to improper fractions before × or ÷

✗ $\frac{2+4}{2+6} = \frac{4}{6}$

✓ $\frac{2+4}{2+6} = \frac{6}{8} = \frac{3}{4}$

cancel factors of the whole top and bottom, not single terms

Fraction rules

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Fraction = division	$\frac{a}{b} = a \div b, \quad n = \frac{n}{1}$
Equivalent fractions	$\frac{a}{b} = \frac{a \times k}{b \times k} = \frac{a \div k}{b \div k}$ same non-zero k on top and bottom; never + or −
Simplify	divide top and bottom by their HCF
Mixed to improper	$p\frac{a}{b} = \frac{pb + a}{b}$
Add / subtract	$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$ plug in; simplify at the end if asked
Multiply	$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$ multiply across
Divide	$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$ flip the second fraction only, then multiply across
Reciprocal	$\frac{a}{b} \to \frac{b}{a}, \qquad \frac{a}{b} \times \frac{b}{a} = 1$
Compare	$\frac{a}{b} > \frac{c}{d} \iff ad > bc$ for positive bottoms
Negative fractions	$-\frac{a}{b} = \frac{-a}{b} = \frac{a}{-b}$
Fraction of an amount	$\frac{a}{b} \text{ of } N = \frac{N}{b} \times a$
Whole and mixed numbers	whole number over 1; mixed number to improper; then the formula

Your turn

Answers are in grey under each group; full working is on the next page.

Simplify and convert

1. Simplify 18/24
2. Write 23/5 as a mixed number
3. Write 3 2/7 as an improper fraction
4. Which is bigger, 5/8 or 7/11?

1. 3/4 2. 4 3/5 3. 23/7 4. 7/11 (5×11 = 55 < 56 = 7×8)

Add and subtract

5. 2/9 + 5/9
6. 3/4 − 2/5
7. 5/6 + 3/8
8. 1/3 − 3/4
9. 4 1/5 − 2 2/3

5. 7/9 6. 7/20 7. 29/24 = 1 5/24 8. −5/12 9. 23/15 = 1 8/15

Multiply and divide

10. 4/9 × 3/8
11. 2 1/3 × 1 1/2
12. 5/6 ÷ 10/9
13. 3/4 ÷ 6
14. 1 1/4 ÷ 2 1/2

10. 1/6 11. 7/2 = 3 1/2 12. 3/4 13. 1/8 14. 1/2

Mixed operations

15. 1/2 + 2/3 × 3/4
16. (1/2 + 2/3) × 3/4
17. 3/5 of 45
18. −2/3 × 3/8

15. 1 16. 7/8 17. 27 18. −1/4

Find the unknown

19. 5/6 = ?/30
20. ? + 1/4 = 7/8
21. ? × 2/3 = 4/9
22. 3/4 of a number is 18. Find it.
23. 2/5 ÷ ? = 4/5

19. 25 20. 5/8 21. 2/3 22. 24 23. 1/2

Answers, worked

1. $\dfrac{18 \div 6}{24 \div 6} = \dfrac{3}{4}$

2. $23 \div 5 = 4 \text{ r } 3 \Rightarrow 4\frac{3}{5}$

3. $\dfrac{3 \times 7 + 2}{7} = \dfrac{23}{7}$

4. $5 \times 11 = 55 < 56 = 7 \times 8 \Rightarrow \dfrac{7}{11}$ is bigger

5. $\dfrac{2 + 5}{9} = \dfrac{7}{9}$

6. $\dfrac{3 \times 5 - 4 \times 2}{4 \times 5} = \dfrac{7}{20}$

7. $\dfrac{5 \times 8 + 6 \times 3}{6 \times 8} = \dfrac{58}{48} = \dfrac{29}{24} = 1\frac{5}{24}$

8. $\dfrac{1 \times 4 - 3 \times 3}{3 \times 4} = -\dfrac{5}{12}$

9. $\dfrac{21}{5} - \dfrac{8}{3} = \dfrac{63 - 40}{15} = \dfrac{23}{15} = 1\frac{8}{15}$

10. $\dfrac{4 \times 3}{9 \times 8} = \dfrac{12}{72} = \dfrac{1}{6}$

11. $\dfrac{7}{3} \times \dfrac{3}{2} = \dfrac{21}{6} = \dfrac{7}{2} = 3\frac{1}{2}$

12. $\dfrac{5}{6} \times \dfrac{9}{10} = \dfrac{45}{60} = \dfrac{3}{4}$

13. $\dfrac{3}{4} \times \dfrac{1}{6} = \dfrac{3}{24} = \dfrac{1}{8}$

14. $\dfrac{5}{4} \div \dfrac{5}{2} = \dfrac{5}{4} \times \dfrac{2}{5} = \dfrac{10}{20} = \dfrac{1}{2}$

15. $\frac{1}{2} + \frac{6}{12} = \frac{1}{2} + \frac{1}{2} = 1$

16. $\frac{7}{6} \times \frac{3}{4} = \frac{21}{24} = \frac{7}{8}$

17. $\frac{45}{5} \times 3 = 9 \times 3 = 27$

18. $-\frac{6}{24} = -\frac{1}{4}$

19. $6 \times 5 = 30 \Rightarrow ? = 5 \times 5 = 25$

20. $? = \frac{7}{8} - \frac{1}{4} = \frac{28 - 8}{32} = \frac{20}{32} = \frac{5}{8}$

21. $? = \frac{4}{9} \div \frac{2}{3} = \frac{4}{9} \times \frac{3}{2} = \frac{12}{18} = \frac{2}{3}$

22. $? = 18 \div \frac{3}{4} = 18 \times \frac{4}{3} = 24$

23. $? = \frac{2}{5} \div \frac{4}{5} = \frac{2}{5} \times \frac{5}{4} = \frac{10}{20} = \frac{1}{2}$

FRACTION RULES CARD

cut it out, keep it next to you

ADD / SUBTRACT

$$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$$

$$\frac{1}{2} + \frac{1}{3} = \frac{1 \times 3 + 2 \times 1}{2 \times 3} = \frac{5}{6}$$

MULTIPLY

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

$$\frac{2}{3} \times \frac{4}{5} = \frac{8}{15}$$

DIVIDE

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

$$\frac{2}{3} \div \frac{4}{5} = \frac{2}{3} \times \frac{5}{4} = \frac{10}{12}$$

SIMPLIFY (OPTIONAL)

$$\frac{a \times k}{b \times k} = \frac{a}{b}$$

$$\frac{10}{12} = \frac{10 \div 2}{12 \div 2} = \frac{5}{6}$$

whole numbers over 1, mixed numbers improper, then plug in